TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATIONS
JC 2/ IP YEAR 6 2017

CANDIDATE NAME	

CLASS	C	G			/	1	6

H2 BIOLOGY

9744/04

Paper 4 Practical

Thursday 24 August 2017
2 hours 30 minutes

Candidates answer on the Question paper
Additional Materials: As listed in the confidential instructions

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/20
2	/21
3	/14
Total	/55

This document consists of **17** printed pages and **1** blank page.

Answer **all** questions

1 Starch grains are found in many plant cells. They are made up of two carbohydrate components, amylose and amylopectin. The glycosidic bonds in amylose are $\alpha(1, 4)$ while those in amylopectin include $\alpha(1, 6)$. Iodine solution stains amylose dark blue and amylopectin red-brown.

You are provided with:

- Unboiled starch: a suspension of starch grains in water
- iodine solution
- Solution **X**
- Spotting tile
- Hand lens

You are required to investigate the action of Solution **X** on starch.

Proceed as follows:

1 Set up a water bath and bring it to boil. This will be required in step **4**. Stir the starch suspension thoroughly. Place a few drops of it on a clean microscope slide. Place the slide on a dark background and examine it using a hand lens.

Record your observations. [1]

White specks/particles observed

2 Move the slide onto a white background. Add a drop of iodine solution to the slide and examine the suspension again.

Record your observations. [1]

Yellow solution with dark blue specks/particles observed.

3 Stir the suspension again and place 6 cm^3 of it into a test-tube labelled “boiled Starch”. At the same time, place 6 cm^3 of Solution **X** in a new test-tube, labelled “boiled **X**”.

4 Place both test-tubes into the boiling water bath for **2** minutes. After this time, remove the test-tubes and cool them under a running tap.

5 Place a few drops of the cooled “boiled Starch” on another microscope slide and add a drop of iodine solution to it. Examine it using a hand lens.

Record your observations. [2]

- **Yellow solution turned dark blue.**
- **uniformly /homogenous**

6 Suggest and explain a possible effect of boiling on the structure of starch grains which would explain the results obtained in step **5.** [2]

1. **increases temperature; increase KE; break intramolecular hydrogen bonds.**
2. **Less compact structure; form hydrogen bonds with water molecules. OWTTE**

*Read the following instructions carefully and prepare a table in **12** to record your observations before starting the investigation.*

7 Label 4 clean and dry vials, **A**, **B, C** and **D**.

8 To each vial, add the following:

Vial **A** : 2 cm^3 unboiled starch + 2 cm^3 unboiled Solution **X**

Vial **B** : 2 cm^3 unboiled starch + 2 cm^3 boiled **X**

Vial **C** : 2 cm^3 boiled starch + 2 cm^3 unboiled Solution **X**

Vial **D** : 2 cm^3 boiled starch + 2 cm^3 boiled **X**

9 Swirl gently to mix the contents in all the vials and leave them on the benchtop for 10 minutes.

10 After 10 minutes, swirl gently to mix the contents in all the vials.

11 Place 2 drops of mixture from each vial in different wells on a spotting tile. Label using the label stickers and paste them at the side of the each well.

12 Add 2 drops of iodine solution to each mixture on the spotting tile. Examine it with a hand lens. Record your observations and conclusions in the space below.

Title: Table of Iodine test on various starch and Solution X mixtures

Samples	**Observations (1/2m each)**	**Conclusions (1/2m each)**
A	**Yellow solution with dark blue particles**	**Starch present**
B	**Yellow solution with dark blue particles**	**Starch present**
C	**Yellow solution turned red brown**	**Amylopectin present**
D	**Yellow solution turned dark blue**	**Starch present**

[5]

13 Using the reagents provided, determine the biomolecule(s) present in Solution **X**.

(a) Observation and conclusion from Biuret test: [1]

1. **blue to violet /light purple; protein is present.**

(b) Observation and conclusion from ethanol emulsion test: [1]

1. **Homogeneous solution was formed with ethanol. Solution remained homogeneous when water was added.**
2. **Lipid absent.**

14 Based on the results obtained in steps **12** and **13**, suggest the identity of Solution **X** and its effect on **unboiled** and **boiled** starch. [3]

1. **Amylase**
2. **Able to hydrolyse glycosidic bonds of the amylose but unable to hydrolysed the amylopectin chains.**

3. **Unable to hydrolyse the amylose chains in unboiled starch**

15 One way to increase the confidence in the conclusions of this investigation would be to repeat the experiment several times.

Describe **two** other modifications to the method that would increase the confidence in the conclusions, and explain how these modifications would achieve this. [4]

1. **Fluctuation of temperature may affect enzymatic reactions thus use thermostatically-controlled water bath at 30°C.**
2. **Volume of iodine solution used is not standardised, thus use 0.2 ml iodine solution for the iodine test.**
3. **AVP**

[Total: 20]

2 Climate change has implications on the physiological processes of plants and animals, including their coping and survival strategies.

During this question you will require access to a microscope and slide **E**.

Slide **E** shows stained sections of leaves from different plants, including Plant **X** which is found in arid habitats. You are **not** expected to have seen this specimen before.

Proceed as follows:

1 Fig. 2.1 shows the outline of the leaf of Plant **X**. Position the slide so that the section is seen as shown in Fig. 2.1. Examine the epidermis on the upper side of the leaf (side **A**) under high-power objective lens.

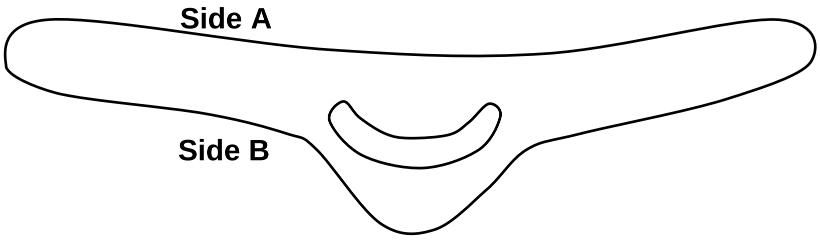
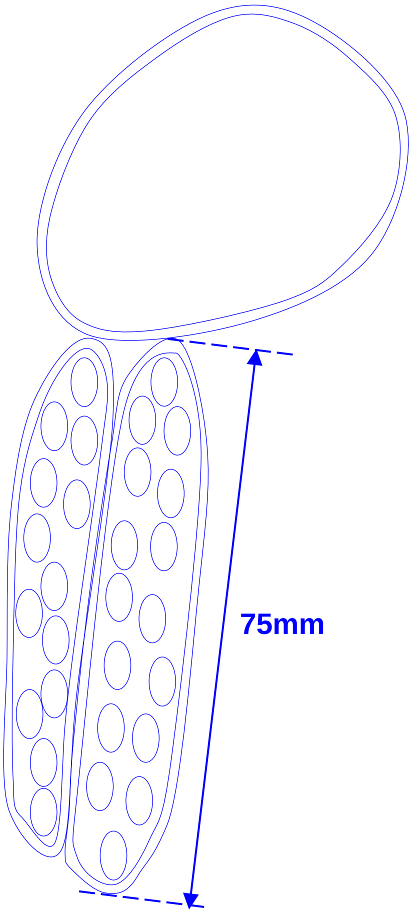


Fig. 2.1

(a) In the space below, make a large, detailed drawing of **one** typical epidermal cell on side **A** and **two** cells attached to it which lie *immediately* internal to it. Labels are **not** required. Calculate the magnification of your drawing.

Detailed drawing of an epidermal cell and two neighbouring cells of Plant X (T.S., 60X)

[4]



Magnification: $\frac{75\ mm}{33 \times 1.67\ \mu m} = 1361X$

(b) Carefully examine the epidermis on side **B** of the leaf for a unique structure that is not present on side **A**.

Relate the role of this structure to the plant's adaptation to the environment. [1]

1. **Stomatal crypts with trichomes (hair-like projections) are present/ OWTTE to trap moisture and reduce water loss in arid habitats.**

The increase in ambient temperature can increase the rate of loss of water in plants. In order to investigate the effect of water loss in various types of plants, an experimental set-up shown in Fig. 2.2 was used to measure the loss in mass of a leaf.

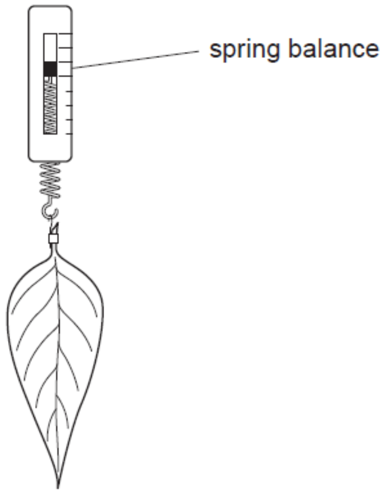


Fig. 2.2

Table 2.1 shows the results of this preliminary investigation.

Table 2.1

Sample	Loss in mass/ g per day			
	Upper side covered with wax		**Lower side covered with wax**	
	Species P	**Species Q**	**Species P**	**Species Q**
1	1.75	1.32	0.75	0.85
2	1.45	1.07	0.85	0.63
3	1.55	1.18	0.75	0.79
4	1.54	1.50	0.95	0.88
5	1.66	1.07	0.75	0.72
Total/ g	7.95	6.14	4.05	3.87
Average loss in mass/ g	1.59	1.23	0.81	0.77
Standard deviation	0.12	0.18	0.09	0.10

(c) With reference to the information given and Table 2.1, identify whether Plant **X** is Species **P** or **Q**. Explain your answer. [3]

1. **Species Q.**

2. **Species Q has SMALLER average mass loss per day through BOTH sides of the leaves (with figures quoted).**

3. **The adaptations of Species Q allow it to minimise water loss in arid (dry) habitat.**

(d) To support your answer in **(c)**, perform a statistical test to determine if the average loss in mass is significantly different between plant species **P** and **Q**. Show your working clearly.

The formulae and probability tables for two statistical tests are given below:

χ^2 – Test formula: $\chi^2 = \sum \frac{(O - E)^2}{E}$

χ^2 – Distribution Table

Degrees of freedom	**Probability, p**				
	0.10	***0.05***	***0.02***	***0.01***	***0.001***
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

***t* – Test formula:** $t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$

***t*- table**

Degrees of freedom	**Probability**				
One-tailed t-test	**0.100**	**0.050**	**0.025**	**0.010**	**0.005**
Two-tailed t-test	**0.200**	**0.100**	**0.050**	**0.020**	**0.010**
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012

14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

Null Hypothesis: There is **NO significant difference** between the average loss in mass of plant species **P** and **Q**.

Alternative Hypothesis: There is **significant difference** between the average loss in mass of plant species **P** and **Q**.

As **water is mainly lost** through the **lower epidermis**, the results for the **upper side covered** was used.

$$t = \frac{|1.23-1.59|}{\sqrt{\frac{0.18^2}{5}+\frac{0.12^2}{5}}} = 3.721$$

Degrees of freedom = (5+5)–2 = 8

1. For **8** degrees of freedom,
2. the calculated *t* value of **3.721** is **more than 2.306**,
3. therefore the **p-value** is **less than 0.05**.
4. The difference in means of the 2 samples is **statistically significant**, and **not due** to **chance**.
5. **Reject** null hypothesis.
6. Hence, **the mean average mass loss per day for Species Q** was **different from** that of species **P**.

[5]

Another experiment (Fig. 2.3) was conducted to investigate the effect of temperature on the rate of water loss in different species of plants. Table 2.2 shows the results of this experiment for plants from Species **P** and Species **Q**.

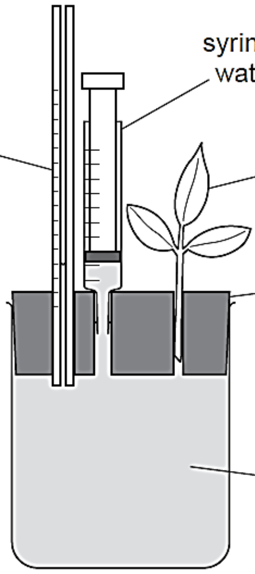


Fig. 2.3

Table 2.2

Temperature/ °C	**Distance moved by water/ mm**					
	Species P			**Species Q**		
	Trial 1	**Trial 2**	**Trial 3**	**Trial 1**	**Trial 2**	**Trial 3**
22	14	15	16	13	11	12
24	28	28	29	16	15	16
26	38	41	42	22	22	21
28	50	52	62	25	24	22
30	63	64	62	29	29	31

(e) Process the data in Table 2.2 and present it clearly in the space below.

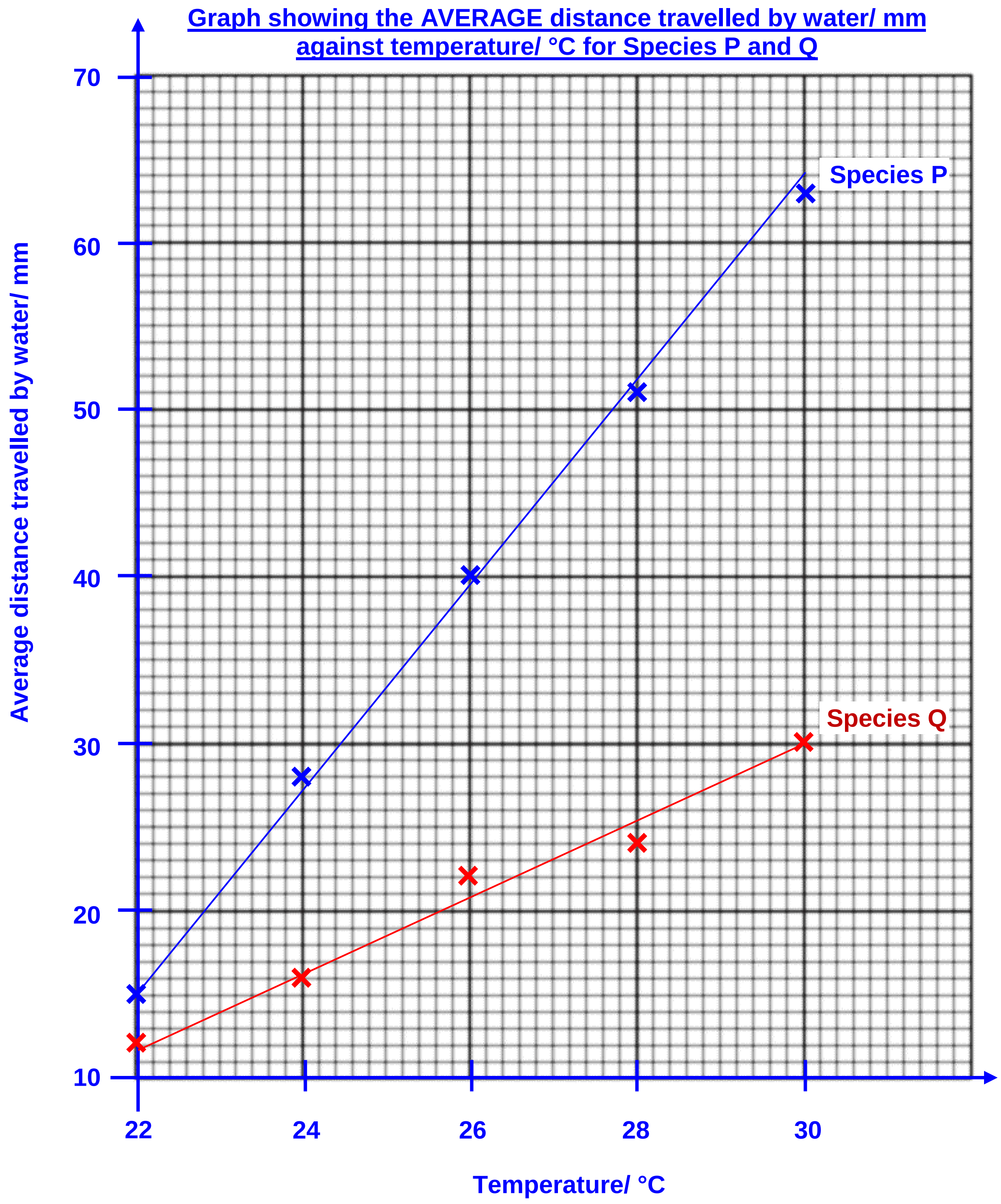
Table showing AVERAGE distance moved by water/ mm at different temperatures/ °C

Temperature/ °C	**Average distance moved by water/ mm**	
	Plant P	**Plant Q**
22	$\frac{14+15+16}{3} = 15$	$\frac{13+11+12}{3} = 12$
24	$\frac{28+28+29}{3} = 28$	$\frac{16+15+16}{3} = 16$
26	$\frac{38+41+42}{3} = 40$	$\frac{22+22+21}{3} = 22$
28	$\frac{50+52}{2} = 51$	$\frac{25+24+22}{3} = 24$
30	$\frac{63+64+62}{3} = 63$	$\frac{29+29+31}{3} = 30$

[3]

(f) Use the grid below to display your results from **(e)**.

[5]



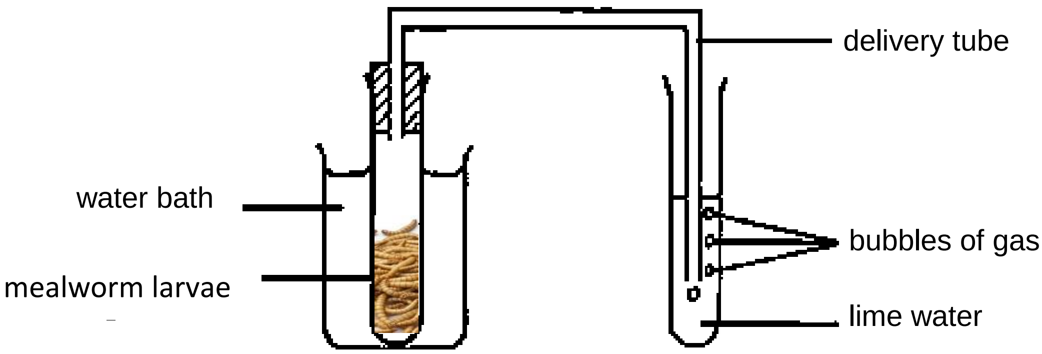
[Total: 21]

3 Insects are among groups of organisms that are most affected by climate change because climatic factors has a direct influence on their development, survival, and reproduction. Moreover, insects have short generation time and high reproductive rate, as such they can respond quicker to climate change.

Mealworm larvae are cold-blooded organisms, thus changes in the environment can affect their rate of movement or activity. The larvae are found naturally inside logs and underneath the bark of dead trees so as to hide from their predators. Thus, they will quickly move away in response to light. Heat can also cause them to become stressed and delay their development into adults, or even die.

A student suggested that temperature and light intensity affect the mealworm's survival.

Modify the set up below to compare the effects of temperature and light intensity (high and low) on the rate of respiration in the mealworm larvae.



You must use the following apparatus:

- mealworm larvae
- lime water
- bench lamp with 30W and 90W bulb

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagrams, if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

A: <u>VARIABLES AND CONTROLLED VARIABLES</u>
1. Independent variable: temperature and light intensity
2. Dependent variable: average number of bubbles produced in five minutes.

Controlled variables
1. Mass of organisms used
    - It affects the volume of carbon dioxide released / rate of respiration.
    - Using a weighing balance, standardize the total mass of the organisms used for each reaction set-up at 20g.

2. Distance of light source from organism
    - It affects the volume of carbon dioxide released / rate of respiration.
    - Using a ruler, standardize the distance of the light source from the boiling tube containing the organism and at 10cm.

3. Duration of experiment
    - It affects the volume of gas measured
    - A digital stopwatch/ timer is used to ensure the duration for respiration is kept constant at 5 minutes.

B: METHOD
1. Set up a thermostatically-controlled water bath at 10°C.
2. Set up a bench lamp with 30W bulb 10cm away from the boiling tube.
3. Place 20g of mealworm larvae inside the boiling tube.
4. Incubate the boiling tube containing the larvae in the water bath for at least five minutes to equilibrate.
5. Attach the delivery tube using a rubber bung and ensure that it is air-tight, and put the other end of the delivery tube into the test tube containing the limewater.
6. Start the digital stopwatch/ timer and time for five minutes.
7. Record the number of bubbles formed in the lime water in the period of five minutes.
8. Repeat steps 1 to 7 to obtain a total of three readings (triplicates) using fresh samples of mealworm larvae and limewater and determine the average volume of bubbles formed.
9. Replace fresh air/oxygen supply in the boiling tube with each repeat.
10. Repeat steps 1 to 9 for at least 4 more temperatures (20, 30, 40, 50°C)
11. Repeat steps 1 to 10, using bench lamp with 90W bulb.
12. Repeat the entire experiment (steps 1 to 11) twice, using the different mealworms and fresh samples of limewater.

C: CONTROL EXPERIMENT
To show that the bubbles produced is due to the effect of respiration in mealworm larvae, a control is set up. Steps 1 to 7 are performed with the same setup, but with only glass beads of the same mass.

D: SAFETY PRECAUTIONS

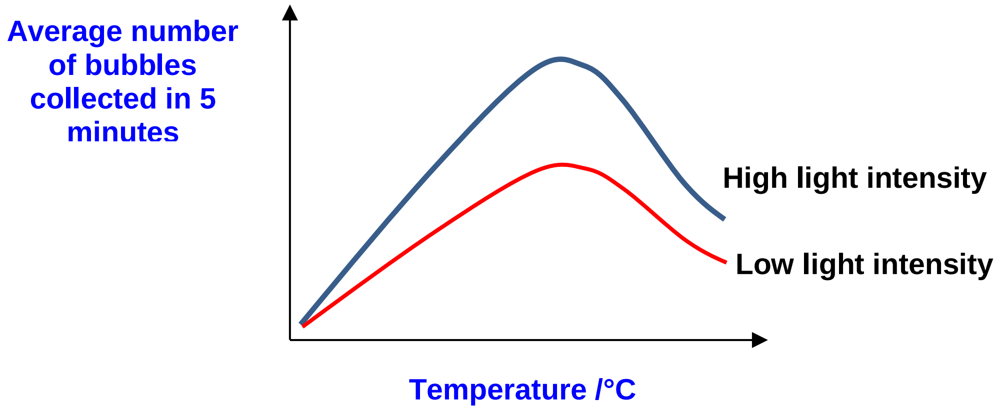
RISK	PRECAUTION
1. <u>Limewater</u> is an irritant	<u>Wear safety goggles</u> and <u>gloves</u> to avoid contact with <u>eyes</u> and <u>skin</u>

E: RECORDING OF RESULTS

Table showing the AVERAGE number of bubbles formed in 5 minutes at different temperatures for low and high light intensities

Temperature /°C	Number of bubbles formed in 5 minutes							
	Low light intensity (30W)				High light intensity (90W)			
	Reading 1	Reading 2	Reading 3	Average	Reading 1	Reading 2	Reading 3	Average
10								
20								
30								
40								
50								

Graph of AVERAGE number of bubbles collected in 5 mins against different temperatures/ °C at different light intensities



1. As temperature increases, the rate of respiration increases, more carbon dioxide is released.
2. Therefore, number of bubbles released increases.
3. As temperature increased beyond optimum temperature, the number of bubbles released decreases as the enzymes in the larvae start to denature. Thus rate of respiration also decreases.
4. At high light intensity, mealworm larvae will move more as they respond to light, thus rate of respiration will be higher than at low light intensity at any given temperature.